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Pathogenic germline variations and cancer risks in pediatric patients referred for genetic testing

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Next-generation sequencing technologies have been widely applied in diagnosing genetic disorders in pediatric patients. However, the cancer predisposition and tumor characteristics in individuals who have germline pathogenic variants remain unclear. We analyzed exome sequencing data from 75,602 pediatric patients referred for genetic testing between January 2016 and January 2025, tracking cancer as a secondary finding. The most common reasons for genetic testing were symptoms related to the nervous system, metabolic disorders and immune dysfunction. Among 110,692 variants of 139 tumor susceptibility genes, we identified 501 (456 single-nucleotide variants, 45 copy number variations, 0.45%) pathogenic or likely pathogenic (P/LP) and 3,848 (3,650 single-nucleotide variants, 198 copy number variations, 3.5%) variants of uncertain significance leaning toward likely pathogenic variants. Of 411 patients with tumors (203 with preexisting tumors and 208 with new tumors diagnosed during follow-up), 134 (32.6%) harbored causative germline P/LP variants in genes such as *NF1* (13.1%), *TSC2* (5.8%), *RB1* (4.6%) and *WT1* (3.2%). Critically, prospective follow-up of 64,187 patients without initial tumors revealed a significantly higher incidence of malignant tumors in those carrying P/LP variants (3.23 per 1,000 person-years) compared with those with variants of uncertain significance leaning toward likely pathogenic or other variants (0.236 and 0.272 per 1,000 person-years, respectively). These findings underscore the importance of proactive genetic counseling and surveillance for pediatric patients with pathogenic germline variants.

Genetic factors strongly influence cancer risk[1–5]. Germline variants, which are heritable DNA alterations in germ cells, are important contributors to cancer development[6]. A study of childhood cancer incidence sponsored by the International Agency for Research on Cancer, including 153 registries from 62 countries between 2001 and 2010, reported approximately 400,000 cases of tumors in children and adolescents aged 0–19 years, with an observed increasing trend[7]. Next-generation sequencing (NGS) has revealed that roughly 10% of pediatric cancer patients harbor germline variants[8–10], with commonly mutated genes including *TP53*, *APC* and *NF1*. Current practice primarily focuses on detecting cancer-related germline variants in pediatric patients already diagnosed with cancer. However, this approach may overlook and underestimate the prevalence of these variants in children without cancer[11]. Evidence suggests that single cancer germline variant screening for newborns is a cost-effective method for identifying carriers of cancer-associated germline variants in both cancer-affected and unaffected children[12].

NGS technology, capable of sequencing billions of DNA templates simultaneously, has become a cornerstone for diagnosing pediatric genetic disorders, enabling the identification of numerous

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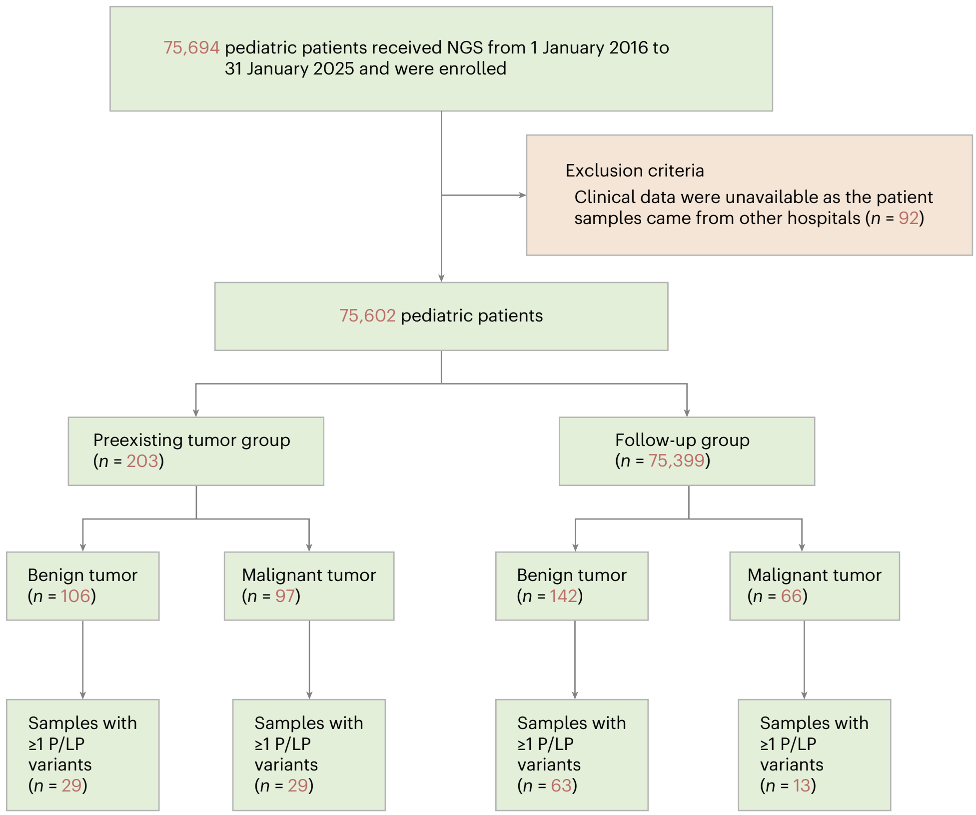

Fig. 1 | Study flowchart. A total of 75,694 pediatric patients received next-generation sequencing between 1 January 2016 and 31 January 2025 and were enrolled. After exclusion of 92 patients whose clinical data were unavailable, 75,602 patients were included and divided into a preexisting tumor group and a follow-up group. The numbers of benign and malignant tumors and samples with at least one pathogenic or likely pathogenic variant are shown for each group.

tumor-predisposing variants[13]. The interpretation of disease-causing variants heavily relies on the guideline established by the American College of Medical Genetics and Genomics (ACMG)[14]. Typically, only variants directly related to the patient's clinical manifestations are reported, while unrelated variants are excluded to avoid unnecessary confusion, following ethical principles. In some cases, laboratories may report secondary findings of other pathogenic and likely pathogenic (P/LP) cancer germline variants unrelated to the patient's symptoms[15]. Critically, the cancer risk for children with germline variants remains poorly understood, particularly owing to the lack of longitudinal studies tracking timed outcomes. These outcomes are essential for interpreting secondary genetic findings, as they provide clinically actionable insights into long-term risks, thereby directly informing counseling strategies and follow-up care recommendations. Thus, it is worth exploring the underlying cancer risk in children with pathogenic germline variants.

Since 2016, we have systematically collected NGS data from pediatric patients referred for diverse clinical indications to construct our study cohort. As of 31 January 2025, over 75,000 pediatric patients have undergone sequencing; specifically, we conducted follow-up on cancer occurrence in patients with non-tumor indications via medical records and telephone interviews. The goal of our study was to profile the underlying cancer risk as well as the tumor characteristics in children who have pathogenic germline variants, providing critical information for performing genetic counseling and early prevention measures.

Results

General characteristics

Between 1 January 2016 and 31 January 2025, a total of 75,694 children underwent clinical genetic testing (Fig. 1). Excluding 92 cases lacking detailed clinical information, we analyzed 75,602 cases with NGS data, focusing on genetic variations in 139 tumor susceptibility genes and the associated long-term cancer predisposition (Supplementary Table 1). The individuals included 28,911 (38.2%) females and 46,691 (61.8%) males. Clinical exome sequencing (CES) tested 70.0% of the individuals, while whole-exome sequencing (WES) tested 30.0% (Extended Data Table 1). The participants were divided into three age groups: 53.5% were newborns (≤28 days), 19.2% were 28 days to 1 year old and 27.3% were 1 year old to 3 years old. The symptoms of patients related to various systems were categorized using Human Phenotype Ontology (HPO) terms presented in Table 1 and Extended Data Fig. 1. The most commonly involved systems were the nervous system, metabolism, digestive system and immune system; only 203 (0.3%) patients were diagnosed with tumors at the time of genetic testing (preexisting tumor group). Of the other 75,399 patients free of tumor manifestations at the time of genetic testing (follow-up group), we obtained follow-up data for 85.1% (64,187/75,399). Subsequently, 208 new tumor cases were diagnosed.

Tumor features

The 203 patients of the preexisting tumor group included 95 females (47%) and 108 males (53%). The age distribution was 38.4% newborns, 23.6% infants (28 days to 1 year) and 37.9% young children (1–3 years). Of these cases, 97 (47.8%) were malignant tumors, while 106 (52.2%) were benign tumors. The most common malignant tumors included Wilms tumor (33/97, 34.0%), retinoblastoma (22/97, 22.7%), neuroblastoma (7/97, 7.2%) and hepatoblastoma (7/97, 7.2%). The most prevalent benign tumors were teratoma (35/106, 33.0%), tuberous sclerosis-related hamartoma (26/106, 24.5%) and neurofibromas (13/106, 12.3%).

Out of the 208 tumor cases from the follow-up group, 113 were male (54.3%) and 95 were female (45.7%). Of these patients, 52.4% (109/208) were older than 6 years, 36.0% (75/208) were between 3 years

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Table 1 | The clinical characteristics and individual symptoms of the cohort

Characteristics	*n*=75,602
Sex	
Female	28,911 (38.2%)
Male	46,691 (61.8%)
Genetic testing age	
≤28 days	40,441 (53.5%)
28 days to 1 year	14,505 (19.2%)
1 year to 3 years	20,656 (27.3%)
Clinical symptoms[a]	
Nervous system	36,374 (48.1%)
Metabolism	23,159 (30.6%)
Digestive	22,700 (30.0%)
Immune	20,324 (26.9%)
Liver	16,012 (21,2%)
Respiratory system	15,949 (21.1%)
Skin	14,538 (19.2%)
Jaundice	14,421 (19.1%)
Cardiovascular system	13,700 (18.1%)
Short stature	11,626 (15%)
Blood	10,188 (13.5%)
Urinary	6,526 (8.6%)
Endocrine	4,398 (5.8%)
Skeletal	4,237 (5.6%)
Maxillofacial deformity	3,752 (5.0%)
Eye	2,163 (2.9%)
Genital system	2,050 (2.7%)
Polydactyly	1,853 (2.5%)
Ear	1,718 (2.3%)
Tumor malignant	97 (0.1%)
Tumor benign	106 (0.1%)

[a]One case may involve more than one system.

Table 2 | Tumors that were already diagnosed at the time of genetic testing (preexisting tumor group) and new tumor cases that appeared from the follow-up group

Characteristics	Preexisting tumor group (*n*=203)	New tumor cases (*n*=208)	*P* value[a]
Sex			0.8
Female	95 (47%)	95 (46%)	
Male	108 (53%)	113 (54%)	
Genetic testing age			0.003
≤28 days	78 (38%)	114 (55%)	
28 days to 1 year	48 (24%)	31 (15%)	
1 year to 3 years	77 (38%)	63 (30%)	
Tumor diagnosis age			<0.001
≤28 days	78 (38%)	0 (0%)	
28 days to 1 year	52 (26%)	5 (2.4%)	
1 year to 3 years	73 (36%)	19 (9.1%)	
3 years to 6 years	0 (0%)	75 (36%)	
>6 years	0 (0%)	109 (52%)	
Variation group			0.017
P/LP	58 (29%)	76 (37%)	
VUS-LP	27 (13%)	40 (19%)	
Other	118 (58%)	92 (44%)	
Tumor type			<0.001
Benign tumor	106 (52%)	142 (68%)	
Malignant tumor	97 (48%)	66 (32%)	

[a]Pearson's chi-squared test. Tumor diagnosis age: $P=1\times10^{-70}$; tumor type: $P=8.8\times10^{-4}$.

and 6 years, and 11.5% (24/208) were less than 3 years. Among them, 66 (31.7%) had malignant tumors, including 26 Wilms tumor, 6 hematological malignancies (3 leukemias, 2 lymphomas and 1 lymphangiosarcoma), 3 hepatoblastomas, 3 retinoblastomas and others. By contrast, 142 (68.3%) had benign tumors, with teratoma (35/142, 24.6%) and tuberous sclerosis-related hamartoma (26/106, 24.5%) being the most common benign tumors. As expected, the proportion of benign tumors was significantly higher in new tumor cases from the follow-up group compared with patients with the preexisting tumor group (68.3% versus 52.2%, $P < 0.001$; Table 2).

Germline variations of pediatric cancer-predisposition genes

In addition to single-nucleotide variants (SNVs), copy number variations (CNVs) affecting 139 genes were extracted from the total samples. A total of 110,692 variants across 139 genes were analyzed, which led to the identification of 456 P/LP and 3,650 variants of uncertain significance leaning toward likely pathogenic (VUS-LP) SNVs, and 45 P/LP and 198 VUS-LP CNVs. Among all 411 tumor cases (203 cases from the preexisting tumor group and 208 cases from the follow-up group), 134 patients (32.6%) had P/LP variants. Of the remaining 277 patients, 76 (18.5%) had VUS-LP variants, while 201 (48.9%) carried neither.

Among patients in the preexisting tumor group, 58 (28.6%) carried P/LP germline variants. Of these, 29 had malignant tumors, primarily involving 19 patients with *RB1* gene variations, 7 patients with *WT1* variations and 3 additional patients each with *SDHC*, *BRAF* and *NRAS* gene variations. The remaining 29 carriers presented with benign tumors, which included 18 patients with *TSC2* gene variations, 5 patients with *NF1* gene variations and 6 additional patients with gene variations in the *TSC1*, *PTEN*, *PMS2*, *MSH2*, *MSH6* and *FLCN* genes, respectively. Furthermore, 27 (13.3%) patients carried VUS-LP variants, while 118 (58.1%) did not have any P/LP or VUS-LP variants (Extended Data Table 2 and Supplementary Table 2).

Among the 64,187 patients we obtained follow-up data from the follow-up group, 1,187 individuals carried P/LP variants, 7,987 individuals did not have P/LP variants but had VUS-LP variants, and 55,013 had neither P/LP nor VUS-LP variants. In the subgroup of 208 newly diagnosed tumor cases, 76 (36.5%) patients were identified as having P/LP germline variants, 40 (19.2%) had VUS-LP variants and 92 (44.2%) had no P/LP or VUS-LP variants (Extended Data Table 2 and Supplementary Table 3).

Within the 76 carriers, we identified 78 distinct P/LP germline variants. The P/LP variants were most frequently found in *NF1* (49 cases), followed by *TSC2* (6), *WT1* (6), *PTPN11* (3) and *TSC1* (3). Notably, these 76 patients accounted for 38.9% (49/126), 6.8% (6/88), 42.9% (6/14), 18.8% (3/16) and 4.7% (3/64) of all P/LP variant carriers for the respective genes in the entire follow-up cohort. Moreover, single occurrences were identified in *APC*, *EXT1*, *GATA2*, *MSH6*, *NF2*, *PALB2*, *PTEN*, *RET*, *SUFU*, *TSHR* and *VHL*. Specifically, two P/LP germline variants were found in case S59052 (*NF2* and *TSHR*) and in case S11325 (*NF1* and *PALB2*). Notably, 13 patients with identified P/LP variants developed malignant tumors: 6 associated with *WT1* variants, 2 with *PTPN11* variants and 1 each associated with variants in *APC*, *GATA2*, *MSH6*, *RET* and *SUFU* (Extended Data Table 2).

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Table 3 | Thirteen malignant tumors diagnosed in follow-up

Sample ID	Sex	Genetic testing age	Clinical symptoms during genetic testing	Gene and variation	Tumor onset age	Diagnosed tumor
S900	Male	2 years	Cerebral palsy, delayed development and epilepsy	PTPN11_chr12_112888199_C_T	3 years and 1 month	Leukemia
S2815	Female	5 days	Jaundice	WT1_chr11_32456245_C_A	5 months	Wilms tumor (both kidneys)
S12784	Male	2 years and 3 months	Seizures and systemic lupus erythematosus	MSH6_chr2_48030639_AC_A	5 years and 11 months	Malignant cerebellar tumor[a]
S14409	Female	1 year and 2 months	Hyperimmunoglobulin E syndrome	PTPN11_chr12_112926888_G_C	5 years and 8 months	Leukemia
S20018	Female	2 days	Premature birth	WT1_chr11_32413566_G_A	1 year and 10 months	Wilms tumor (right kidney)
S20351	Male	2 years and 1 month	Cryptorchidism	WT1_chr11_32414250_C_G	2 years and 1 month	Gonadoblastoma
S30798	Male	3 months	Delayed development	SUFU_chr10_104263999_TC_T	6 months	Malignant cerebellar tumor[b]
S48216	Male	2 days	Maternal syphilis and jaundice	APC_chr5_112174489_ACAAT_A	2 years and 8 months	hepatoblastoma
S48193	Female	5 months	Inflammatory bowel disease	RET_chr10_43617416_T_C	4 years and 9 months	Wilms tumor (right kidney)
S51519	Male	3 days	Neonatal pneumonia	GATA2_chr3_128200724_G_A	5 years and 1 month	Leukemia
S51674	Male	1 year	Abnormal liver function	WT1_chr11_32456559_CG_C	1 year and 9 months	Wilms tumor (both kidneys)
S55400	Female	1 year and 1 month	Patent ductus arteriosus	WT1_chr11_32456267_G_A	1 year and 11 months	Wilms tumor (both kidneys)
S55553	Male	1 year and 1 month	Agranulocytosis	WT1_chr11_32413578_G_A	2 years	Wilms tumor (left kidneys)

[a]This case was pathologically confirmed as medulloblastoma. [b]At 6 months of age, a cerebellar tumor was detected by MRI. The patient did not undergo surgery and died in the emergency department at the age of 2 years and 6 months.

The correlation between phenotype and germline genotype in patients with tumors

In the 411 tumor cases, a total of 227 P/LP or VUS-LP germline variants across 50 pediatric cancer-predisposition genes were involved. Among these genes, the *NF1* gene was observed in 83 patients (20.2%), the *TSC2* gene in 26 patients (6.3%), the *RB1* gene in 19 patients (4.6%) and the *WT1* gene in 13 patients (3.2%). Meanwhile, the *RET* and *SEC23B* were present in 7 cases, *MSH6* appeared in 5 cases, *TSC1* appeared in 5 cases and *PTPN11*, *PTEN*, *SDHB* and *BLM* each had 3 cases.

Therefore, the most common genes associated with benign tumors were *NF1*, causing type 1 neurofibromatosis, and *TSC2*, leading to type 2 tuberous sclerosis. Conversely, common early-onset malignant tumors with pathogenic germline variations included retinoblastoma caused by the *RB1* gene and Wilms tumors caused by variations in the *WT1* gene. In the 13 follow-up cases of malignant tumors, there were 6 Wilms tumors, 3 leukemias, 2 malignant cerebellar tumors, 1 hepatoblastoma and 1 gonadoblastoma; the associated gene variations are listed in Table 3. Among them, patient S12784 underwent WES at 2 years and 3 months for seizures and systemic lupus erythematosus. He was diagnosed with malignant cerebellar medulloblastoma at 5 years and 11 months, with the tissue classified as Sonic Hedgehog-activated type and having a TP53 mutation. In addition to a P/LP variant identified in *MSH6*, three VUS-LP variants were also found in the *BLM*, *MSH6* and *XPC* genes.

The inheritance patterns of the 50 genes included autosomal dominant (AD, $n = 24$), autosomal recessive (AR, $n = 10$), AD/AR ($n = 13$) and unclassified (NA, $n = 3$). No autosomal recessive gene was identified as the cause of the tumor; all P/LP or VUS-LP variants were included in the analysis of tumor risk. Among the 134 tumor patients with a cause of genetic P/LP variation, all 23 genes showed dominant inheritance. Among the genes inherited in an autosomal dominant manner, the variation in the *RB1* gene has the highest penetrance in retinoblastoma (90.5%, 19/21). Conversely, the variation in the *WT1* gene shows the lowest penetrance in Wilms tumors (52.0%, 13/25).

Underlying tumor risk analysis by a survival strategy

The overall tumor proportion is 0.5% (411/75,602), while the malignant tumor proportion is 0.2% (163/75,602) in our samples. The association between germline variants and tumor risk was investigated among the 64,187 patients sequenced for non-tumor indications via survival analysis. Patients with P/LP variants (18.9 per 1,000 person-years) or VUS-LP variants (1.58 per 1,000 person-years) show higher tumor incidence rates than patients with other variants (0.532 per 1,000 person-years).

Crucially, regarding malignant tumors, the incidence rate was markedly higher in the P/LP group (3.23 per 1,000 person-years) than in the VUS-LP variants (0.236 per 1,000 person-years) or other variants (0.272 per 1,000 person-years). Consistent with this, the P/LP group shows significantly poorer tumor-free probability ($P < 0.0001$) and higher cumulative incidence ($P < 0.0001$) compared with the remaining groups. Further subgroup analysis revealed that the other variant group showed superior outcomes compared with the VUS-LP group, with higher tumor-free probability ($P < 0.0001$) and lower cumulative incidence ($P < 0.05$) (Fig. 2).

Discussion

Between 2018 and 2020, the reported incidence of pediatric cancer in China was 122.86 per million (0.012%)[16], with a germline variant carrier rate of 8.5% in pediatric cancer patients[8]. However, existing studies have primarily focused on children with cancer[8,10], leaving the clinical penetrance and long-term tumorigenic risk of these variants in the asymptomatic or non-cancer pediatric population largely unexplored. In this study, we leveraged data from a large-scale genomic study to investigate the underlying cancer risk as well as tumor characteristics in children who have pathogenic germline variants. We found that pathogenic germline variants in cancer-predisposing genes were identified in 1.9% (1,404/75,399) of children. In the follow-up group of 64,187 children, the malignant tumor incidence was 0.1% (66 cases).

In the 411 tumor cases, the tumor patterns of the two groups differed. In patients who had previously been diagnosed with a tumor before the genetic test group, the most prevalent benign tumors were teratoma, tuberous sclerosis-related hamartoma and neurofibromas. These benign tumors were common in pediatric patients, and malformations or seizures were the major clinical symptoms prompting genetic tests. The spectrum of cancer types in this study differs from those in previous cancer cohorts[8,10]. In our study, the most common malignant tumors included Wilms tumor, retinoblastoma, neuroblastoma and hepatoblastoma. These four malignant tumors were the most common cancers occurring in early life. However, in the follow-up group, the leading benign tumors among the 208

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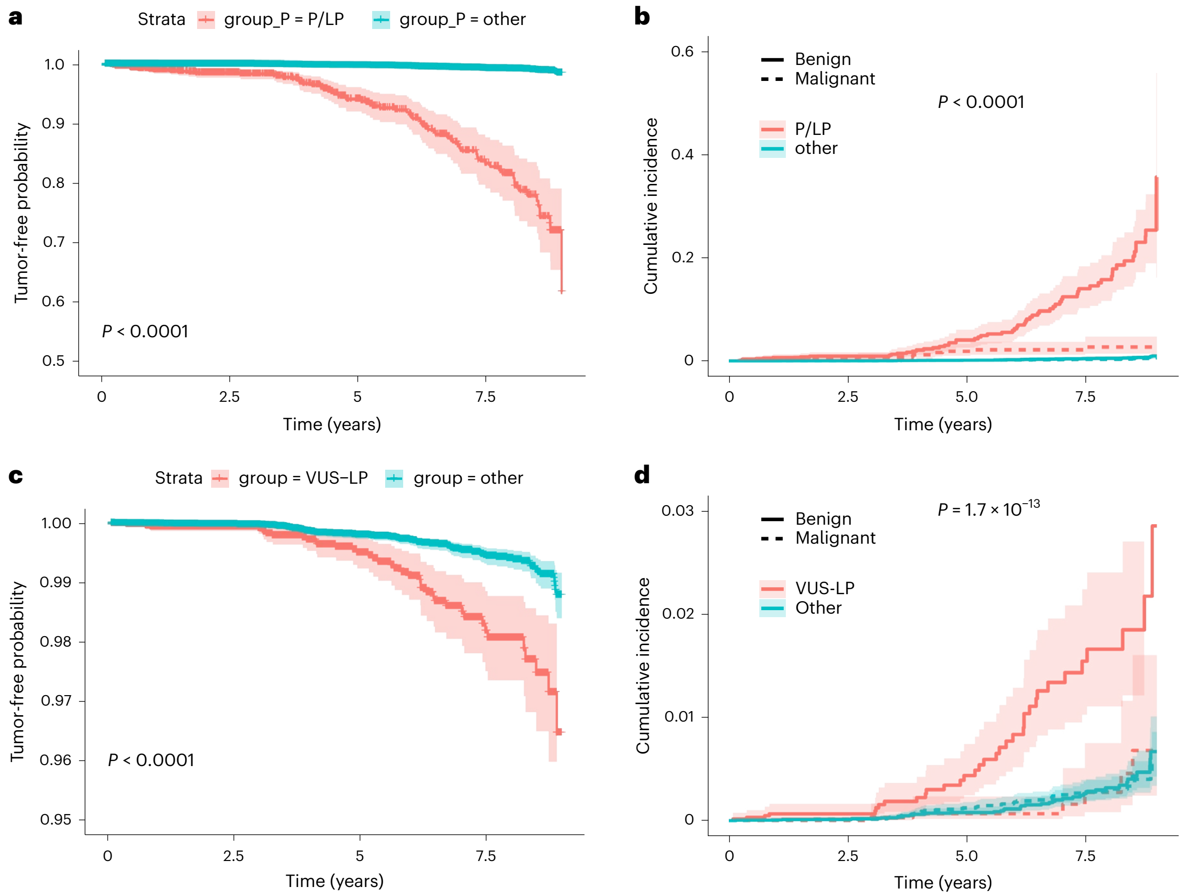

Fig. 2 | Tumor risk in the follow-up group by survival strategy. a, The tumor-free probability decreased over the course of follow-up years in individuals with P/LP variants of cancer-predisposition genes ($P = 2.3 \times 10^{-261}$). **b**, The cumulative incidence increased over the course of follow-up years in individuals with P/LP variants, particularly after the age of 5 ($P = 1.8 \times 10^{-9}$). **c**, The tumor-free probability decreased over the course of follow-up years in individuals with VUS-LP variants of cancer-predisposition genes ($P = 6.3 \times 10^{-101}$). **d**, The cumulative incidence increased over the course of follow-up years in individuals with VUS-LP variants, particularly after the age of 5 ($P = 5.2 \times 10^{-13}$). Shaded areas represent 95% pointwise confidence intervals for the cumulative incidence functions. For **a** and **b**, between-group comparisons were performed using the log-rank test (a nonparametric method for comparing survival distributions), with a two-sided alternative hypothesis and no adjustment for multiple comparisons. The *P* value shown in the figure corresponds to the result of the log-rank test. For **c** and **d**, between-group comparisons were performed using the competing risk model, with a two-sided alternative hypothesis and no adjustment for multiple comparisons. The *P* values shown in the figure are derived from the Wald test for the grouping variable in the model.

individuals subsequently identified were neurofibromas and tuberous sclerosis-related hamartoma. The proportion of benign tumors is higher in the follow-up group than in the preexisting tumor group. The difference in tumor type distribution between the two groups can be explained by their distinct identification pathways. In the patients with tumors at genetic testing, malignancies are more likely to be detected because their symptoms prompt immediate clinical evaluation. By contrast, the follow-up group better reflects the natural history of tumor development, in which benign tumors are more common and may only become apparent over time. For malignant tumors, except for Wilms tumor, the incidence of other blastomas decreased, and the types of cancer varied.

In the follow-up group, there are 18 cases with pathogenic variants in the *WT1* gene. Around 33.3% of these individuals had tumor diagnoses, including five Wilms tumors and one gonadoblastoma. The variants in these six cases were all truncating variations. Wilms tumor is a childhood embryonic tumor of the kidney characterized by genetic heterogeneity and (epi)genetic susceptibility factors[17]. The *WT1* gene accounts for approximately 20% of Wilms tumors. P/LP variations in the *WT1* gene may lead to chronic kidney disease, sex reversal and/or abnormalities in the genitourinary system; patients with truncating mutations have the highest risk of developing type 1 Wilms tumor[18]. In addition, at the last follow-up, 140 individuals with *NF1* and 33 individuals with *TSC2* gene P/LP variants have not experienced tumor onset, despite having received genetic diagnoses. General follow-up and genetic counseling will be provided to these families. Interestingly, while pathogenic variants in DNA repair genes, such as *PMS2*, *TP53* and *BRCA*, dominate previous pediatric cancer cohorts[8,10], our study reveals *RB1*, *TSC2* and *WT1* as the most prevalent predisposition genes. Notably, brain tumors were underrepresented in our cohort, which may have led to an under-detection of cancer-predisposition genes associated with brain tumors (such as *TP53* and *NF1*). Therefore, these findings should be interpreted within this context. Nonetheless, our findings still suggest that in unselected pediatric cohorts, genes associated with organ-specific tumors or developmental syndromes may be more prevalent than those linked to high-penetrance familial cancer syndromes, necessitating more tailored genetic management.

In clinical genetic tests, ethical guidelines often restrict the disclosure of cancer risks associated with germline variants in children to mitigate anxiety. Genetic counselors have a critical role in providing comprehensive explanations about genetic diseases, evaluating the risk of developing other conditions and providing guidance on appropriate interventions[19]. Our findings revealed that 17.5% (208/1,187) of patients with pathogenic germline variants had tumors during follow-up, underscoring the importance of genetic counseling for these cancer-predisposition genes. Given that our recruited participants comprised a clinical referral cohort of pediatric patients presenting with diverse, serious non-cancer phenotypes, our findings are best interpreted within this specific context. Therefore, while the germline variant constitutes the primary and necessary genetic insult, the complex physiological milieu of serious non-cancer diseases may actively promote oncogenesis through established mechanisms. The chronic inflammation, immune dysregulation or aberrant tissue repair processes characteristic of many such conditions could synergistically create a fertile 'tumor-promoting environment'[20]. This environment may not only increase genomic instability but also theoretically accelerate the acquisition of the necessary somatic 'second hits' in accordance with Knudson's hypothesis[21], thereby hastening the progression from genetic predisposition to overt malignancy.

As newborn genetic screening becomes increasingly widespread[22,23], careful interpretation of cancer germline variants will be essential. Collaboration between genetic counselors and healthcare professionals is needed to accurately assess the potential risk and clinical relevance of these variants. Some pediatric cancer-predisposition genes were the cause of an inherited syndrome. Pediatric patients may present with clinical symptoms, and a genetic test will aid in genetic diagnoses. Genetic counseling will be provided to the patient's family for tumor surveillance. On the contrary, pediatric patients with cancer should also be examined by a clinical geneticist or a pediatrician skilled in clinical morphology to determine whether the patients have a malformation syndrome[24]. Notably, while the clinical utility of these cancer-predisposition mutations is unproven, future large cohort studies are needed.

This study has several limitations. First, we only analyze 139 pediatric cancer-predisposition genes; a substantial portion of genes potentially associated with cancer development may have been overlooked. Second, the study exclusively recruited pediatric patients, limiting our ability to comprehensively assess cancer incidence, especially for cancers that develop later in life. In addition, the relatively short follow-up period may have led to an underestimation of cancer incidence, particularly for tumors with a later peak age such as pediatric brain tumors. As most children have not yet reached the higher-risk age range, this also explains the lower number of brain tumor cases observed, in contrast to earlier-onset malignancies such as retinoblastoma. Moreover, epigenetic predisposing factors[17] and known intrinsic risk factors, such as birth weight and parental age, have been linked to pediatric cancer[25]; due to limited data, these factors were not evaluated in this study. We will continue to monitor tumor prevalence in our study.

In conclusion, pathogenic germline variants in pediatric cancer-predisposition genes were identified in 1.9% (1,404/75,399) of children in our follow-up group and have a higher tumor risk, with 0.1% (66/64,187) showing malignant tumor phenotypes. Our findings highlight the potential to enhance genetic counseling and promote early, personalized interventions to prevent cancer. Considering the limitations of our study, we advocate for future research to include broader investigations of cancer-associated genes and longitudinal follow-up in future studies.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41591-026-04423-5.

References

1. Samadder, N. J., Baffy, N., Giridhar, K. V., Couch, F. J. & Riegert-Johnson, D. Hereditary cancer syndromes—a primer on diagnosis and management, part 2: gastrointestinal cancer syndromes. *Mayo Clin. Proc.* **94**, 1099–1116 (2019).
2. Chen, X. et al. Risk of syndrome-associated cancers among first-degree relatives of patients with pancreatic ductal adenocarcinoma with pathogenic or likely pathogenic germline variants. *JAMA Oncol.* **9**, 955–961 (2023).
3. Wen, H. et al. Prevalence and landscape of pathogenic or likely pathogenic germline variants and their association with somatic phenotype in unselected Chinese patients with gynecologic cancers. *JAMA Netw. Open* **6**, e2326437 (2023).
4. Huang, J. -q & Luo, H. -y Impact of genome and epigenome on intratumor heterogeneity in colorectal cancer. *MedComm Futur. Med.* **2**, e34 (2023).
5. Zheng, Y., Huang, J., Xu, Y. & Luo, H.-Y. Current progress in chimeric antigen receptor T-cell therapy for malignant tumors. *MedComm Futur. Med.* **3**, e79 (2024).
6. Schrader, K. A. et al. Germline variants in targeted tumor sequencing using matched normal DNA. *JAMA Oncol.* **2**, 104–111 (2016).
7. Steliarova-Foucher, E. et al. International incidence of childhood cancer, 2001–10: a population-based registry study. *Lancet Oncol.* **18**, 719–731 (2017).
8. Zhang, J. et al. Germline mutations in predisposition genes in pediatric cancer. *N. Engl. J. Med.* **373**, 2336–2346 (2015).
9. Parsons, D. W. et al. Diagnostic yield of clinical tumor and germline whole-exome sequencing for children with solid tumors. *JAMA Oncol.* **2**, 616–624 (2016).
10. Grobner, S. N. et al. The landscape of genomic alterations across childhood cancers. *Nature* **555**, 321–327 (2018).
11. Fiala, E. M. et al. Prospective pan-cancer germline testing using MSK-IMPACT informs clinical translation in 751 patients with pediatric solid tumors. *Nat. Cancer* **2**, 357–365 (2021).
12. Kunst, N. et al. Population-based newborn screening for germline TP53 variants: clinical benefits, cost-effectiveness, and value of further research. *J. Natl Cancer Inst.* **114**, 722–731 (2022).
13. Sahoo, O. S. et al. Role of next-generation sequencing in revolutionizing healthcare for cancer management. *MedComm Futur. Med.* **3**, e70001 (2024).
14. Richards, S. et al. Standards and guidelines for the interpretation of sequence variants: a joint consensus recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology. *Genet. Med.* **17**, 405–424 (2015).
15. Miller, D. T. et al. ACMG SF v3.1 list for reporting of secondary findings in clinical exome and genome sequencing: a policy statement of the American College of Medical Genetics and Genomics (ACMG). *Genet. Med.* **24**, 1407–1414 (2022).
16. Ni, X. et al. Socioeconomic inequalities in cancer incidence and access to health services among children and adolescents in China: a cross-sectional study. *Lancet* **400**, 1020–1032 (2022).
17. Hol, J. A. et al. Prevalence of (epi)genetic predisposing factors in a 5-year unselected national Wilms tumor cohort: a comprehensive clinical and genomic characterization. *J. Clin. Oncol.* **40**, 1892–1902 (2022).
18. Lipska, B. S. et al. Genotype–phenotype associations in WT1 glomerulopathy. *Kidney Int.* **85**, 1169–1178 (2014).
19. Catapano, F. et al. The role of the genetic counsellor in the multidisciplinary team: the perception of geneticists in Europe. *Eur. J. Hum. Genet.* **30**, 1432–1438 (2022).
20. Rahman, N. Realizing the promise of cancer predisposition genes. *Nature* **505**, 302–308 (2014).
21. Knudson, A. G. Two genetic hits (more or less) to cancer. *Nat. Rev. Cancer* **1**, 157–162 (2001).
22. King, J. S. & Smith, M. E. Whole-genome screening of newborns? The constitutional boundaries of state newborn screening programs. *Pediatrics* **137**, S8–S15 (2016).
23. Minten, T. et al. Data-driven consideration of genetic disorders for global genomic newborn screening programs. *Genet. Med.* **27**, 101443 (2025).

24. Merks, J. H., Caron, H. N. & Hennekam, R. C. High incidence of malformation syndromes in a series of 1,073 children with cancer. *Am. J. Med. Genet. A* **134A**, 132–143 (2005).
25. Spector, L. G., Pankratz, N. & Marcotte, E. L. Genetic and nongenetic risk factors for childhood cancer. *Pediatr. Clin. North Am.* **62**, 11–25 (2015).

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the International Consortium in Digital Twins in Healthcare and Medicine

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Methods

Ethics approval

The study was approved by the Ethics Committee of Guangzhou Women and Children's Medical Center (approval number 2025111A01) and the Ethics Committee of the Children's Hospital of Fudan University (approval number 2024-286), and all analyses were conducted using de-identified data. Informed consent was obtained from patients' guardians. For the publication of findings, all data presented in this paper have been rigorously aggregated and anonymized to prevent patient re-identification. All of the participants' guardians agreed to the publication of the data including indirect identifiers.

Statistics and reproducibility

Our study initially included 75,694 pediatric patients (<3 years old) referred for genetic testing of suspected hereditary disorders between 2016 and 2025. A total of 92 patients were excluded from the study owing to unavailability of their clinical data. The remaining 75,602 patients were stratified into a 'preexisting tumor group' ($n = 203$), encompassing patients with preexisting tumor diagnosis at genetic testing, and a 'follow-up group' ($n = 75{,}399$), consisting of patients without tumors at enrollment who were prospectively followed for cancer occurrence. No statistical method was used to predetermine sample size. Data combined retrospective clinical records with prospective follow-up data collected from the follow-up group. No data were excluded from the analyses. The aim is to re-analyze cancer risk in pediatric patients focusing on a pediatric cancer-predisposition gene panel. Patients were prospectively followed from the time of genetic testing until either tumor onset or 31 May 2025, whichever occurred first. Retrospective clinical data from birth to the time of NGS testing were retrieved from the electronic medical record system.

Tumors

Clinical data and follow-up information were collected through electronic medical records and telephone interviews. The final follow-up was performed between 1 December 2024 and 31 May 2025. The diagnosis of the tumor was extracted from the clinical diagnosis in the medical record. Tumors were categorized as benign or malignant following established criteria from JAMA Oncology[26].

Pediatric cancer-predisposition genes

We selected inherited cancer syndrome genes, childhood cancer susceptibility genes and genetic disease genes increasing the risk of childhood tumors based on previous studies' protocols[27] and the ACMG newly published guidance for reporting secondary findings (SF) in the context of clinical exome and genome sequencing (the ACMG SF v.3.1) list for reporting SF[28]. From a list of 605 candidate genes, 16 inherited cancer syndrome genes (*APC*, *CDH1*, *CDKN1B*, *DICER1*, *MEN1*, *MLH1*, *MSH2*, *MSH6*, *PMS2*, *PTCH1*, *PTEN*, *RB1*, *RET*, *TP53*, *VHL* and *WT1*) and 26 genes recommended by ACMG SF v.3.1 were selected. Additional genes were selected based on two criteria: genes described in the Online Mendelian Inheritance in Man and had disease-causing mutation (DM) records in the Human Gene Mutation Database professional version (HGMD professional). In total, 139 pediatric high-risk cancer-predisposition genes were selected (Supplementary Table 1).

NGS and clinical information dataset

The sequencing was performed using CES or WES. The NGS methods have been described previously[29,30]. Briefly, samples were extracted from whole blood with a QIAamp DNA Blood Mini kit (Qiagen) following the manufacturer's protocol. The quality and quantity of the DNA samples were assessed using a NanoDrop 2000 spectrophotometer (Thermo Fisher Scientific). DNA fragments were enriched for CES (including 2,742 genes) or WES with the Agilent ClearSeq Inherited Disease panel kit (Agilent Technologies), the Twist Core Exome with patches (Twist Bioscience) or the Agilent SureSelect XT Human All Exon V5 kit, along with the xGen Exome Research Panel v.2 (Integrated DNA Technologies). Sequencing was conducted on the Illumina HiSeq 2000/2500, Illumina X Ten/Illumina NextSeq/Illumina Novaseq 6000 platform. The average on-target sequencing depth for CES was approximately 200×; for WES, it was approximately 120×.

Data analysis was conducted using an in-house pipeline (https://zenodo.org/records/16892762; Supplementary Table 2)[31]. Raw sequencing reads were aligned to the reference human genome assembly (NCBI build 37/hg19) with Burrows-Wheeler Aligner (v.0.5.9-r16) to get BAM files. Variant calling was performed either with the Verita Trekker platform (Berry Genomics) or with Sentieon[32] to get VCF files.

SNV analysis was conducted using an in-house computational pipeline (https://zenodo.org/records/16892762 (file: src/SNV_filter.pl); Supplementary Table 2)[31] to pre-process and filter raw variants based on the following criteria: (1) left align all variants, (2) remove low-quality (read depth < 10) and deep-intron variants with CADD score < 20 and spliceAI score < 0.2, (3) retrieve public-reported pathogenic variants ('DM/DM? in HGMD or P/LP in ClinVar and allele frequency<0.005 in gnomAD and ExAC' OR 'DM/DM? in HGMD and P/LP in ClinVar and allele frequency<0.01 in gnomAD and ExAC') and (4) annotate variants to genes. The VCF files for the 139 candidate gene panels were extracted from the NGS data of enrolled pediatric patients for the next analysis.

Then, the variants were automated annotated and evaluated using an in-house pipeline called CLEVER (CLinical Evidence-Based Variant Interpreter; https://zenodo.org/records/15534528)[33], developed in accordance with the ACMG guidelines[34], and the latest recommendations from the Clinical Genome Resource (ClinGen) Sequence Variant Interpretation Working Group. Variant annotation was performed using the Ensembl Variant Effect Predictor (VEP), integrating data from public databases including the Genome Aggregation Database (gnomAD), ClinVar and the HGMD, as well as internal databases containing previously diagnosed cases. CLEVER also incorporates state-of-the-art computational functional prediction scores. The aggregated information is then translated into standardized ACMG evidence codes, allowing for evidence upgrading or downgrading based on ClinGen's most recent recommendations. Pathogenicity classification is subsequently assigned to each variant using a Bayesian scoring framework, in line with ACMG guidelines. Finally, candidate pathogenic variants (predicted as P/LP level) were manually reviewed by the experts following the ACMG guidelines. The variant prioritization process was combined with a phenotype-scoring algorithm named PhenoPro[35].

For the CNV calling, exon-level read counts for the captured regions were generated from the BAM file with BedTools[36] with default parameters. We also built a CNV detection pipeline PICNIC[31,37] (https://zenodo.org/records/16892762; Supplementary Table 2). Raw coverage files from the same sequencing batch were merged, coverage for chrX of male samples was modified and the coverage region was refined according to the sequencing capture. The CNV calling was performed by CANOES[38] and HMZDelFinder[39]. The pathogenicity of CNVs was annotated with reference to the literature and the following genetic databases: UCSC Genome Browser (http://genome.ucsc.edu/), DECIPHER (https://decipher.sanger.ac.uk/) and ClinGen (https://clinicalgenome.org/). ClassifyCNV[40] and dbCNV[28] were used to get the predicted pathogenicity for CNVs. Similarly, candidate pathogenic variants (predicted as P/LP level) were manually reviewed by the experts with PhenoPro used to assist the variant prioritization by phenotype.

Clinical data, including sex, age, major clinical symptoms, clinical diagnoses and tumor follow-up information, were extracted from medical records. Major clinical features were mapped to the HPO. We then defined the phenotypes of the HPO into 19 major clinical categories based on the mapping table with 17,751 HPO terms.

Data analysis

Individuals were diagnosed with tumors (including benign and malignant) before the genetic diagnosis was grouped as the preexisting tumor group; others were categorized as the follow-up group, and further follow-up data were analyzed. All germline variations from 139 panel genes in the cohort were defined as P/LP, VUS-LP and other (VUS, benign and likely benign) (Supplementary Table 2).

For each individual in the follow-up group, if a tumor event was found in the follow-up data, the time of the first recorded tumor occurrence was noted as the event time. If a patient developed a malignant tumor from a benign tumor, we recorded the first malignant tumor event as the event time. The state of the tumor (benign or malignant) at the time of the event was recorded as the event state. If an individual's tumor occurrence time had not been observed at the last follow-up, we noted the last follow-up time. For all individuals in the follow-up group, we used a survival analysis strategy as follows: First, we divided them into two primary groups for initial comparison: those carrying ≥1 P/LP variants and all others. The tumor-free probability between these two groups was compared using the log-rank test. Next, to specifically evaluate the risk associated with VUS-LP variants, we excluded individuals with P/LP variants and compared the tumor-free probability between those with ≥1 VUS-LP variants and those with neither P/LP nor VUS-LP variants. Then, regarding tumor status (benign or malignant), we conducted a competing risk analysis to examine the impact of these genetic grouping on the development of benign versus malignant tumors. Finally, for the final analysis, we categorized the individuals into three mutually exclusive groups: (1) P/LP carriers, (2) VUS-LP-only carriers (without P/LP) and (3) individuals with neither P/LP nor VUS-LP variants. For these three groups, we calculated the total person-years and the incidence rate per 1,000 individuals for any tumor occurrence and malignant tumor occurrence. The analysis code to reproduce the results for main figures and tables is available via Zenodo at https://zenodo.org/records/19600900 (ref. 41).

Statistical analysis
Categorical variables are presented as frequency counts and proportions and compared with the $\chi 2$ test. $P < 0.05$ was considered statistically significant. Statistical analysis was performed using Stata v.20 (StataCorp).

Reporting summary
Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability
The sequencing data from our study have been deposited securely in the Genome Sequence Archive housed within the National Genomics Data Center at the China National Center for Bioinformation/Beijing Institute of Genomics, Chinese Academy of Sciences. The VCF files of the 139 candidate gene panels of individuals without cancer and raw sequencing data of cancer patients are available for research only at BioProject under accession code PRJCA042078. Restrictions apply to the availability of datasets, which were used with the permission of the participants for the current study. Data access requests should be addressed to W.Z., who will evaluate data access requests with the data access committee for approval. The request must comply with Institutional Review Board requirements and be for research use only, as applicable. Response will be provided in seven working days. Approved researchers will be able to access the data after completing registration and agreeing to a data-sharing agreement through the portal.

Code availability
The annotation tool CLEVER used in this study is available via Zenodo at https://doi.org/10.5281/zenodo.15534528 (ref. 33). The home-made pipeline for SNV and CNV variant calling and annotation from fastq files is available via Zenodo at https://doi.org/10.5281/zenodo.16892762 (ref. 31). The analysis pipeline and code to reproduce the main results are available via Zenodo at https://doi.org/10.5281/zenodo.19600900 (ref. 41).

References

26. Patel, A. Benign vs malignant tumors. *JAMA Oncol.* **6**, 1488 (2020).
27. Jovanović, A. et al. Germline variants in cancer predisposition genes in pediatric patients with central nervous system tumors. *Int. J. Mol. Sci.* **24**, 17387 (2023).
28. Lv, K. et al. dbCNV: deleteriousness-based model to predict pathogenicity of copy number variations. *BMC Genom.* **24**, 131 (2023).
29. Xiao, F. et al. Genetic spectrums and clinical profiles of critically ill neonates with congenital auricular deformity in the China Neonatal Genomes Project. *Hum. Genet.* **142**, 1737–1745 (2023).
30. Wang, H. et al. Genetic architecture in neonatal intensive care unit patients with congenital heart defects: a retrospective study from the China Neonatal Genomes Project. *J. Med. Genet.* **60**, 247–253 (2023).
31. Dong, X. Home-made pipeline for SNV/CNV variant calling and annotation from fastq files. *Zenodo* https://doi.org/10.5281/zenodo.16892762 (2025).
32. Kendig, K. I. et al. Sentieon DNASeq variant calling workflow demonstrates strong computational performance and accuracy. *Front. Genet.* **10**, 736 (2019).
33. Wang, Y. CLEVER: CLinical Evidence-based Variant interpretER. *Zenodo* https://doi.org/10.5281/zenodo.15534528 (2025).
34. Kleinberger, J., Maloney, K. A., Pollin, T. I. & Jeng, L. J. An openly available online tool for implementing the ACMG/AMP standards and guidelines for the interpretation of sequence variants. *Genet. Med.* **18**, 1165 (2016).
35. Li, Z. et al. PhenoPro: a novel toolkit for assisting in the diagnosis of Mendelian disease. *Bioinformatics* **35**, 3559–3566 (2019).
36. Quinlan, A. R. & Hall, I. M. BEDTools: a flexible suite of utilities for comparing genomic features. *Bioinformatics* **26**, 841–842 (2010).
37. Dong, X. et al. Clinical exome sequencing as the first-tier test for diagnosing developmental disorders covering both CNV and SNV: a Chinese cohort. *J. Med. Genet.* **57**, 558–566 (2020).
38. Backenroth, D. et al. CANOES: detecting rare copy number variants from whole exome sequencing data. *Nucleic Acids Res.* **42**, e97 (2014).
39. Du, H. et al. HMZDupFinder: a robust computational approach for detecting intragenic homozygous duplications from exome sequencing data. *Nucleic Acids Res.* **52**, e18 (2024).
40. Gurbich, T. A. & Ilinsky, V. V. ClassifyCNV: a tool for clinical annotation of copy-number variants. *Sci. Rep.* **10**, 20375 (2020).
41. Dong, X. Analysis pipeline and code for 'Pathogenic germline variations and cancer risk in pediatric patients'. *Zenodo* https://doi.org/10.5281/zenodo.19600900 (2026).

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Author contributions

F.L., H.L., K.Z. and W.Z. conceived, designed and supervised the project. Data collection and analysis were performed by H.W., X.D., F.X., G.L., Y.L., M.Q., B.W., Q.N., K.Y., Q.L., Z.Y., L.Y., D.C., L.C., W.K., Q.W., L.T., G.C., L.W., K.L., C.S., S.W., X.X., M.Y., F.L., H.L., K.Z. and W.Z. The paper was written by H.W., X.D., F.X., G.L., F.L., H.L., K.Z. and W.Z. All authors discussed the results and reviewed the paper.

Competing interests

The authors declare no competing interests.

Additional information

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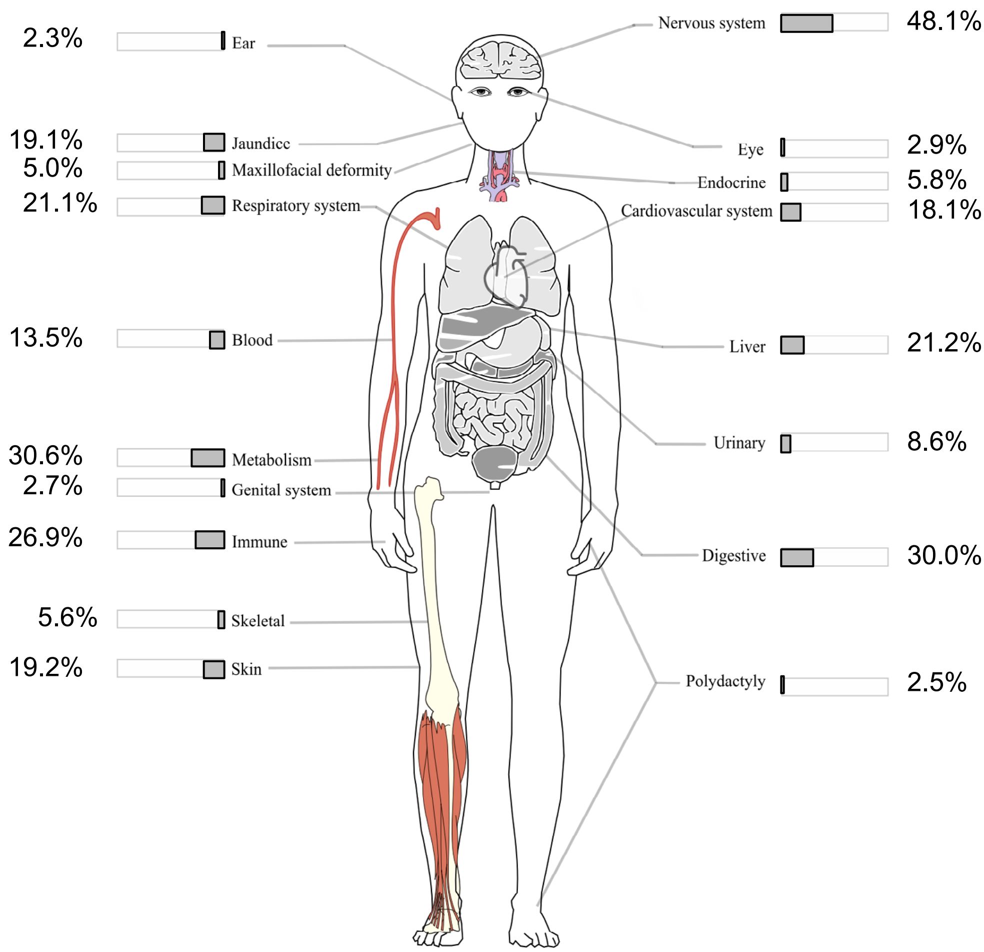

Extended Data Fig. 1 | Distribution of clinical symptoms across organ systems using HPO terms. The symptoms of patients, as summarized in genetic tests related to different systems, were categorized using HPO terms. One case may have more than one HPO term (ranging from 1 to 23 HPO terms). Credit: image adapted from a schematic by ozhank/Openclipart.

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Extended Data Table 1 | NGS data used in this study

Platform of next-generation sequencing	**Capture description of clinical exome sequencing (CES)**	**Number of CES**	**Capture description of whole exome sequencing (WES)**	**Number of WES**
Illumina HiSeq 2500	Agilent ClearSeq Inherited Disease panel kit	105	Agilent SureSelect XT Human All Exon V5 kit	2329
Illumina X ten	Agilent ClearSeq Inherited Disease panel kit	1152	Agilent SureSelect XT Human All Exon V5 kit	5198
Illumina Nextseq &Novaseq	Agilent ClearSeq Inherited Disease panel kit	20294	Agilent SureSelect XT Human All Exon V5 kit	11203
Illumina Novaseq 6000	IDTxGenExome	25823	IDTxGenExomeV2	1578
Illumina Novaseq 6000	Twist cwes	5533	Twist Exome plus	2387
Platform of next-generation sequencing	Capture description of clinical exome sequencing (CES)	Number of CES	Capture description of whole exome sequencing (WES)	Number of WES
Illumina HiSeq 2500	Agilent ClearSeq Inherited Disease panel kit	105	Agilent SureSelect XT Human All Exon V5 kit	2329
Illumina X ten	Agilent ClearSeq Inherited Disease panel kit	1152	Agilent SureSelect XT Human All Exon V5 kit	5198
Illumina Nextseq &Novaseq	Agilent ClearSeq Inherited Disease panel kit	20294	Agilent SureSelect XT Human All Exon V5 kit	11203
Illumina Novaseq 6000	IDTxGenExome	25823	IDTxGenExomeV2	1578
Illumina Novaseq 6000	Twist cwes	5533	Twist Exome plus	2387

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Extended Data Table 2 | Tumor types and genetic features of the P/LP variants carriers

	Tumor types	Pre-existing tumor group (n = 58)	New cases from the follow-up group (n = 76)*
Malignant tumors	Retinoblastoma	*RB1* (n = 19)	/
	Wilms tumor	*WT1* (n = 7)	*WT1* (n = 5), *RET* (n = 1)
	Others	*NRAS, SDHC,* and *BRAF* (n = 1)	*PTPN11* (n = 2), *APC*, *MSH6*, *GATA2*, *WT1*, and *SUFU* (n =1)
Benign tumors	Neurofibromas	*NF1* (n = 5)	*NF1* (n = 45), *PALB2* (n = 1), *TSC2* (n = 1)
	Cardiac rhabdomyoma	*TSC2* (n = 3), *FLCN* (n = 1)	*NF1* (n = 1)
	Tuberous sclerosis	*TSC2* (n = 15), *MSH2* (n = 1), TSC1 (n = 1)	*TSC1* (n = 3), *TSC2* (n = 5), *NF1* (n =2)
	Teratoma	*PMS2* (n = 1), *MSH6* (n = 1)	/
	Others	*PTEN* (n = 1)	*PTEN*, *PTPN11, EXT1, NF1*, *NF2, TSHR,* and *VHL* (n = 1)

* S11325 and S59052 carry two genes

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Reporting Summary

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Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	No special software or code was used to collect the data
Data analysis	Statistical analysis was performed using Stata version 20 (StataCorp; College Station, TX, USA). The annotation tool CLEVER used in this study is deposited on Zenodo (DOI: 10.5281/zenodo.15534528). The home-made pipeline for SNV/CNV variant calling and annotation from fastq files is deposited on Zenodo (DOI: 10.5281/zenodo.16892762). The analysis pipeline and code to reproduce main results is deposited on Zenodo (DOI: 10.5281/zenodo.17032133). Besides, the code is available from the developer upon reasonable request made to the corresponding author.

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Reporting on sex and gender	We used "sex" in our manuscript because we considered the biological attributes.
Reporting on race, ethnicity, or other socially relevant groupings	Information on race, ethnicity or other socially relevant groupings was not available for all included datasets.
Population characteristics	We established an integrated cohort by combining retrospective clinical features and existing NGS databases with prospective follow-up data, comprising a large cohort of pediatric patients who underwent genetic testing for suspected hereditary diseases at the Clinical Genetics Laboratory of the Children’s Hospital of Fudan University or Guangzhou Women and Children's Medical Center between January 1, 2016, and January 31, 2025. Clinical data and follow-up information were collected through electronic medical records and telephone interviews. Totally, 75602 cases were enrolled. And clinical data and genetic informations were used for analyzed. For the publication of findings, all data presented in this manuscript have been rigorously aggregated and anonymized to prevent patient re-identification. All of the participants agreed to publish their results anonymously. Sex analysis was not carried out in this study. The primary objective of this study was to define the spectrum and penetrance of germline variants in a large, unselected pediatric clinical cohort and to estimate the associated population-level cancer risk. The study was not specifically powered or designed to test hypotheses regarding sex-based biological differences in cancer predisposition. Therefore, all primary analyses (e.g., variant prevalence, incidence rates, survival analyses) were performed on the aggregate cohort.
Recruitment	All participants provided written informed consents by guardians before enrollment. Participants were recruited as a consecutive, unselected clinical referral cohort comprising all pediatric patients (n=75,602) referred for genetic testing of suspected hereditary disorders at the participating centers between 2016 and 2025. This design inherently introduces a referral or ascertainment bias, as the cohort represents children with clinical indications severe enough to warrant genetic testing, not the general pediatric population. This bias likely leads to an overestimation of the prevalence of pathogenic germline variants and the associated cancer risks compared to the general population, but accurately reflects the risk within this high-risk clinical subgroup.
Ethics oversight	The study was approved by the Ethics Committee of Guangzhou Women and Children's Medical Center (approval numbers: 2025111A01) and the Ethics Committee of the Children’s Hospital of Fudan University (approval number: 2024-286)

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	Totally, 75,602 cases were enrolled in this study. A formal a priori sample size calculation was not performed. This was a retrospective analysis of a consecutive, unselected clinical cohort comprising all pediatric patients referred for genetic testing at the participating centers between 2016 and 2023, constituting the entire available population during that period. The exceptionally large sample size provides substantial statistical power to detect the effects of interest (e.g., differences in tumor incidence rates) and ensures precise estimates for variant prevalence and associated risks.
Data exclusions	The exclusion criteria were patients who lack of available clinical data.

Replication	Not relevant. The present study is an observational analysis of a unique, real-world clinical cohort. Its core data, the patient population, their inherent germline variants, and the occurrence of tumors, are intrinsic clinical facts, not the outcome of a controlled experiment, and therefore cannot be replicated or verified through repeated experimentation.
Randomization	Not relevant. This study was an observational cohort study, not an intervention-based experiment. Therefore, participants were not "allocated" into groups. They were categorized based on their inherent clinical status at the time of genetic testing (forming the "pre-existing tumor group" vs. the "follow-up group") or based on the objective results of their genetic testing (e.g., carriers of P/LP variants). As this grouping reflects real-world clinical presentation and laboratory findings, random allocation was neither performed nor applicable.
Blinding	All data were de-identified to remove any patient-related information. A formal blinding procedure was not applicable or relevant in this study, which was an observational, retrospective-prospective cohort study that analyzed pre-existing, objective data: germline variant classifications were determined by standardized pipelines and ACMG guidelines, and tumor diagnoses were based on clinical records, not on investigator assessment. Since the core data points (genetic status and cancer occurrence) were not subject to investigator interpretation or measurement bias, blinding to group allocation was not a methodological concern.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☐	☒ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Clinical data

Policy information about clinical studies

All manuscripts should comply with the ICMJE guidelines for publication of clinical research and a completed CONSORT checklist must be included with all submissions.

Clinical trial registration	Our study retrospective analyzed cases with genetic test in Children's hospitals. Given its non-interventional and absence of controlled cases, we did not register this study in ClinicalTrials.gov or any equivalent agency.
Study protocol	Study protocol is available in our previous studies.
Data collection	We established an integrated cohort by combining retrospective clinical features and existing NGS databases with prospective follow-up data, comprising a large cohort of pediatric patients who underwent genetic testing for suspected hereditary diseases at the Clinical Genetics Laboratory of the Children’s Hospital of Fudan University or Guangzhou Women and Children's Medical Center between January 1, 2016, and January 31, 2025. Patients younger than 3 years were included. The exclusion criteria were individual patients who lacked available clinical data. The aim is to re-analyze cancer risk in pediatric patients focusing on a pediatric cancer-predisposition gene panel. Patients were prospectively followed from the time of genetic testing until either tumor onset or May 31, 2025, whichever occurred first. Retrospective clinical data from birth to the time of NGS testing were retrieved from the electronic medical record system.
Outcomes	The main focus of our study was to assess the rate of tumor occurence in patients with germline variants. Additionally, as part of our secondary outcomes, we sought to determine the genetic spectrums in tumor patients.

Plants

Seed stocks	*Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.*
Novel plant genotypes	*Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.*
Authentication	*Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosiacism, off-target gene editing) were examined.*